Stability Analysis

To assess the stability of PDDFNet across different runs, we evaluate the model on five datasets using three additional random seeds (123, 2025, and 3000) and compare the results with those reported in our paper (seed 42). Hyperparameters are kept fixed across runs. Table 1 reports the corresponding MSE and MAE results over four forecasting horizons for each dataset.

Table 1: Results on five datasets over four forecasting horizons under four random seeds. AVG denotes the average across the four forecasting horizons.

Seed		42		123		2025		3000	
Metrics		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	0.3662	0.3856	0.3683	0.3857	0.3691	0.3859	0.3662	0.3857
	192	0.4259	0.4155	0.4264	0.4150	0.4255	0.4155	0.4262	0.4155
	336	0.4502	0.4215	0.4490	0.4212	0.4466	0.4208	0.4476	0.4215
	720	0.4791	0.4547	0.4793	0.4546	0.4799	0.4550	0.4815	0.4561
	AVG	0.4304	0.4193	0.4308	0.4191	0.4303	0.4193	0.4304	0.4197
ETTh2	96	0.2702	0.3223	0.2710	0.3229	0.2701	0.3224	0.2701	0.3225
	192	0.3413	0.3684	0.3417	0.3691	0.3416	0.3689	0.3411	0.3686
	336	0.3362	0.3723	0.3366	0.3723	0.3364	0.3722	0.3364	0.3720
	720	0.4079	0.4238	0.4086	0.4237	0.4053	0.4229	0.4120	0.4252
	AVG	0.3389	0.3717	0.3395	0.3720	0.3384	0.3716	0.3399	0.3721
ETTm1	96	0.3135	0.3444	0.3110	0.3416	0.3109	0.3406	0.3128	0.3431
	192	0.3553	0.3666	0.3552	0.3656	0.3574	0.3670	0.3547	0.3656
	336	0.3732	0.3872	0.3755	0.3892	0.3753	0.3869	0.3734	0.3871
	720	0.4466	0.4255	0.4472	0.4267	0.4493	0.4269	0.4475	0.4282
	AVG	0.3722	0.3809	0.3722	0.3808	0.3732	0.3804	0.3721	0.3810
Weather	96	0.1666	0.2003	0.1656	0.1994	0.1659	0.1991	0.1649	0.2010
	192	0.2103	0.2409	0.2096	0.2413	0.2094	0.2410	0.2108	0.2415
	336	0.2528	0.2788	0.2532	0.2769	0.2530	0.2801	0.2545	0.2792
	720	0.3414	0.3344	0.3407	0.3340	0.3410	0.3331	0.3430	0.3342
	AVG	0.2428	0.2636	0.2423	0.2629	0.2423	0.2633	0.2433	0.2640
Electricity	96	0.1579	0.2443	0.1558	0.2429	0.1560	0.2451	0.1581	0.2433
	192	0.1682	0.2531	0.1679	0.2544	0.1662	0.2559	0.1674	0.2549
	336	0.1834	0.2688	0.1830	0.2671	0.1845	0.2668	0.1834	0.2690
	720	0.2221	0.3015	0.2214	0.3009	0.2231	0.3019	0.2244	0.3020
	AVG	0.1829	0.2669	0.1820	0.2663	0.1825	0.2674	0.1833	0.2673

PDDFNet's performance remains stable across different random seeds, as shown by the results. Both MSE and MAE show minor fluctuations, indicating that the performance improvements reported in our paper are consistent and robust to initialization.